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# High-resolution sequence stratigraphy of a complex, incised valley succession, Cobequid Bay – Salmon River estuary, Bay of Fundy, Canada

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## ABSTRACT

The post-glacial succession in the Cobequid Bay – Salmon River incised valley contains two sequences, the upper one incomplete. The lower sequence contains only highstand system tracts (HST) deposits which accumulated under microtidal, glacio-marine deltaic conditions. The upper sequence contains two, retrogradationally stacked parasequences. The lower one accumulated in a wave-dominated estuarine environment under micro-mesotidal conditions. It belongs to the lowstand system tract (LST) or early transgressive system tract (TST) depending on the timing and location of the lowstand shoreline, and contains a gravel barrier that has been overstepped and preserved with little modification. The upper parasequence accumulated in the modern, macrotidal estuary, and is assignable to the late TST. Recent, net progradation of the fringing marshes indicates that a new HST has begun.

The sequence boundary separating the two sequences was formed by fluvial incision, and perhaps also by subtidal erosion during the relative sea level fall. Additional local erosion by waves and tidal currents occurred during the transgression. The base of the macrotidal sands is a prominent tidal ravinement surface which forms the flooding surface between the backstepping estuarine parasequences. Because fluvial deposition continued throughout the transgression, the fluvial-estuarine contact is diachronous and cannot be used as the transgressive surface. The maximum flooding surface will be difficult to locate in the macrotidal sands, but is more easily identified in the fringing muddy sediments.

These observations indicate that: (1) large incised valleys may contain a compound fill that consists of more than one sequence; (2) relative sea level changes determine the stratal stacking patterns, but local environmental factors control the nature of the facies and surfaces; (3) these surfaces may have complex origins, and commonly become amalgamated; (4) designation of the transgressive surface (and thus the LST) is particularly difficult as many of the prominent surfaces in the valley fill are diachronous facies boundaries; and (5) the transgression of complex topography may cause geologically instantaneous changes in tidal range, due to resonance under particular geographical configurations.

## INTRODUCTION

With the growth of sequence stratigraphy, new emphasis has been placed on understanding the origin and stratigraphical organization of incised valley fills (Dalrymple *et al.*, 1994). Estuarine deposits are a major constituent of these systems, and ancient estuarine successions are being reported with increasing frequency (e.g. Reinson *et al.*, 1988; Van Wagoner *et al.*, 1990; Zaitlin & Schultz, 1990; Ricketts, 1991). Most previous stratigraphical studies of modern, incised valley estuaries have concentrated on micro-mesotidal, wave-dominated systems (*sensu* Dalrymple *et al.*, 1992b; e.g. Demarest & Kraft, 1987; Knebel *et al.*, 1988; Colman *et al.*, 1990; Boyd & Honing, 1992). Among macrotidal estuaries, stratigraphical information exists only for the Gironde River, France (Allen *et al.*, 1970; Allen & Truilhé, 1988; Allen, 1991; Allen & Posamentier, 1993), South Alligator River, Australia (Woodroffe *et al.*, 1989, 1993), Chignecto Bay, Bay of Fundy (Amos & Zaitlin, 1985; Amos *et al.*, 1991), Cobequid Bay and Salmon River (Dalrymple *et al.*, 1990, 1992a) and the Severn River estuary (Allen & Rae, 1988; Allen, 1990). Of these, a sequence stratigraphical interpretation exists only for the Gironde.

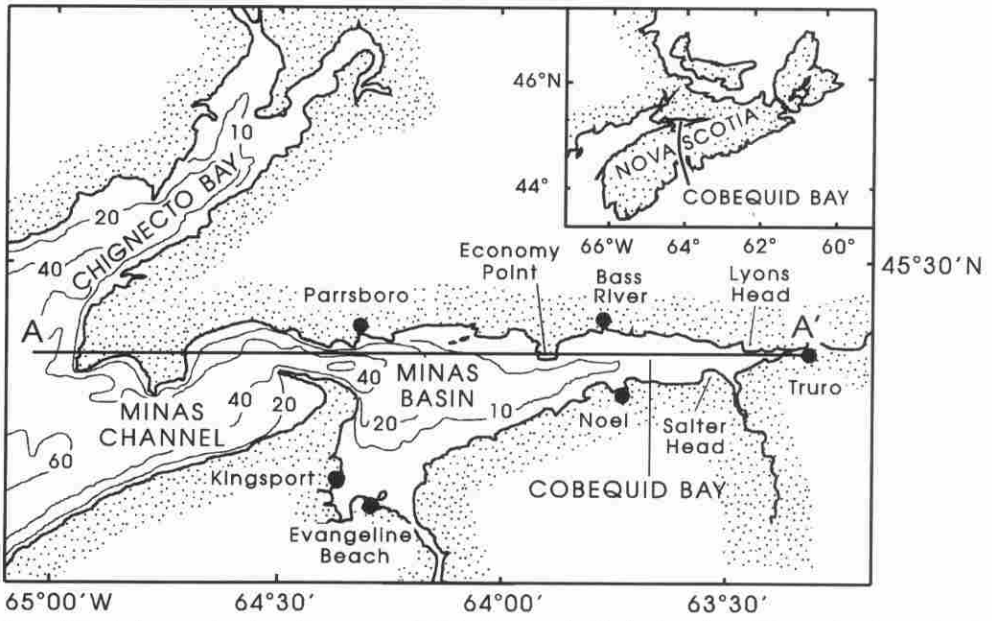


**Fig. 1.** Map of the study area showing place names cited in the text. A–A′: the location of the section shown in Fig. 4.

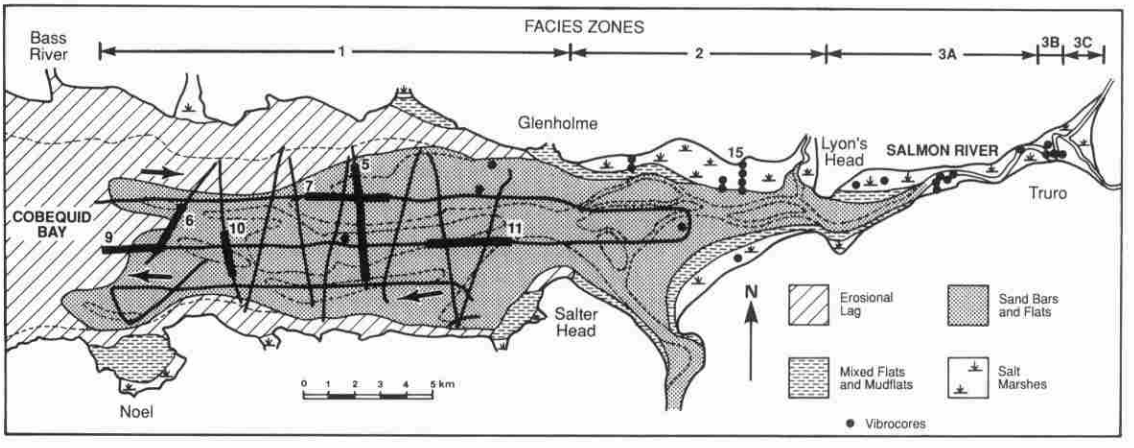


**Fig. 2.** Distribution of facies and facies zones (Dalrymple *et al.*, 1990) within the present day estuary. The locations of vibrocores (dots) and seismic survey track lines (solid lines) are indicated. The heavy bars show the location of the seismic sections shown in the accompanying figure number. The sand bar outline (dashed lines) is based on bathymetric data collected in this study. Arrows: the direction of net sediment transport; the major flood-dominant channel lies along the northern shore of the bay, whereas the main ebb channel runs along the south shore. The bayline, where the (steep) fluvial and (flat) tidally determined coastal plain profiles intersect (Posamentier *et al.*, 1988; Allen & Posamentier, 1993), is situated in facies zone 3B. The tidal limit lies 1–2 km landward of this.

In order to extend our understanding of sedimentation within macrotidal estuaries, we describe here the facies evolution and stratal stacking patterns of the deposits within the Cobequid Bay – Salmon River (CB-SR) valley, Bay of Fundy (Figs 1 & 2). Due to rapid sedimentation, the estuary has passed the Holocene transgressive limit and has begun to prograde (Dalrymple *et al.*, 1990); thus, it can be argued to have entered highstand time (although future events may require modification of this

assignment). As a result, there is a unique opportunity to examine its stratigraphical organization prior to dissection during sea level fall, or modification during renewed sea level rise. We will also discuss how the important, sequence stratigraphical surfaces (sequence boundary, flooding surfaces, ravinement surfaces and transgressive surface) may be expressed within estuaries of this type. A comparison will also be made with other macrotidal estuaries, to assess the generality of the observations.

This paper is an extension of previous work on the CB-SR area (Dalrymple & Zaitlin, 1989; Dalrymple *et al.*, 1990, 1991, 1992a,b), and readers are referred to these papers for more information on the geological history and present day facies. The location and orientation of the CB-SR valley are structurally controlled, and incision of the Triassic bedrock may have begun in the Cretaceous. The most recent period of erosion occurred during Pleistocene glaciation (Stea, 1984). The valley is up to 10 km wide (Fig. 2), and its base is generally <40 m below the present high tide level. The Pleistocene to modern valley fill deposits locally reach 30 m in thickness. At present, 95–99% of the sediment entering the system is derived from areas seaward of the estuary, and is transported headward by tidal currents and waves (Amos & Long, 1980; Dalrymple *et al.*, 1990). Fluvial input is now minimal, but was higher during deglaciation (Hennigar, 1972). During the accumulation of the valley fill deposits, relative sea level initially fell but then rose to its present position. Dramatic changes in tidal range have also occurred (see below).

The sequence stratigraphical terminology used here conforms broadly with the definitions of the Exxon school (eg. Vail *et al.*, 1977; Posamentier *et al.*, 1988, 1992a; Posamentier & Vail, 1988; Van Wagoner *et al.*, 1988, 1990), and particularly as applied by Allen & Posamentier (1993).

## DATA BASE AND METHODOLOGY

This study is based on the integration of two data sets (Fig. 2): a high resolution, reflection seismic survey in the outer and middle estuary, which was conducted during a 5 day period in August 1988; and a set of 35, 7·6 cm (3 inch) diameter vibrocores (maximum penetration 7 m) obtained mainly from the middle and inner estuary. Several vibrocores were collected from the sand bars of the outer estuary, but these did not penetrate deeply enough to ground-truth the seismic data. Twenty-seven $^{14}C$ dates were obtained on wood, organic material and shells from the vibrocores, to determine the chronology of the deposits. Additional evidence for the nature and timing of events is derived from studies conducted elsewhere in the inner Bay of Fundy (e.g. Amos, 1978; Scott & Greenberg, 1983; Amos & Zaitlin, 1985; Amos *et al.*, 1991).

Approximately 100 km of record were collected in the seismic survey, from an area measuring 22 km by 6·5 km (Fig. 2). The energy source was a Nova Scotia Research Foundation 20 tip, 3·5 kHz sparker system, fired once per second with an energy of 100 J. Penetration averaged 15–20 m; throughout, a speed of sound of 1500 m $s^{-1}$ has been assumed for both water and sediment. The receiver consisted of a nine element hydrophone with a centre weighted response. The data were recorded using an EPC recorder with a paper speed of 2·35 cm $min^{-1}$. Positions were determined using Loran-C, with automated annotation of the records every 100 m of track line. Comparison of reflector elevations at the 31 crossover points suggests that most of the position fixes have a precision of ± 10–20 m. The practical, vertical resolution of the seismic data was approximately ± 0·25 ms two way travel time (i.e. ± 0·2 m).

In order to contour the topography of the seismic surfaces, all elevations were standardized to a mean high water elevation of 12·4 m, using hourly, tidal height predictions supplied by the Canadian Hydrographic Service. Topographical and isopach maps were then constructed using values measured at 200 m intervals along each survey line: thus, each map is based on more than 600 data points.

## SEA LEVEL AND TIDAL RANGE HISTORY

Deglaciation of the Minas Basin area occurred 14 000–15 000 yr BP (Amos, 1978; Stea & Wightman, 1987). Eustatic sea level rise coupled with glacio-isostatic depression of the area produced a local highstand at this time (Fig. 3). Due to subsequent differential rebound (in response to spatially variable ice loads), the associated shoreline is now 30–40 m above present sea level in Chignecto Bay, but decreases to the east, intersecting the present high tide level in Cobequid Bay (Wightman & Cooke, 1978; Fig. 4). As the ice retreated, outwash plains and deltas were deposited in the Salmon River valley at Truro (Hennigar, 1972) and along the north shore of Cobequid Bay and the Minas Basin (Wightman,

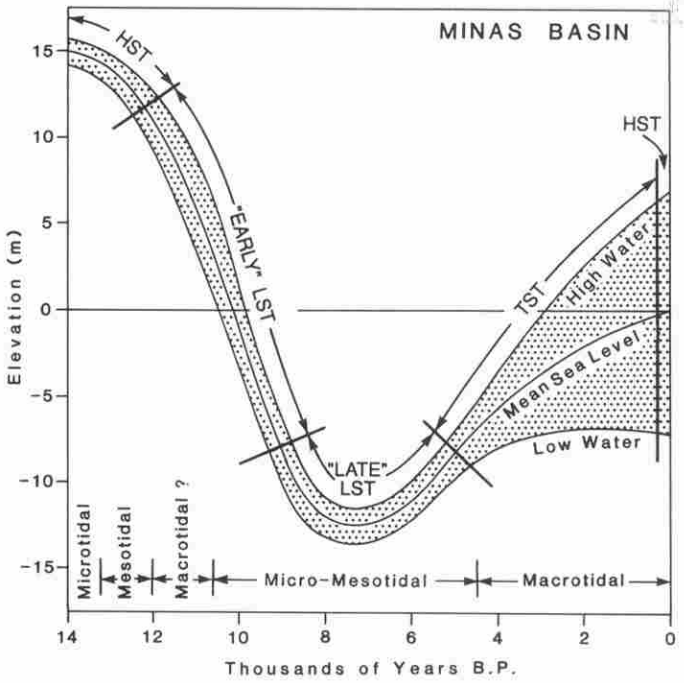


**Fig. 3.** History of relative sea level (considering only eustasy and isostasy/tectonics) in the Minas Basin, based on data from the Parrsboro-Kingsport-Evangeline Beach area (Fig. 1; Dalrymple *et al.*, 1992a), together with the duration of the inferred systems tracts. Early LST and Late LST correspond with the Lowstand Fan and Lowstand Wedge Systems Tracts of Van Wagoner *et al.* (1988). If sediment deposition is considered, relative sea level would now be falling in the Cobequid Bay – Salmon River estuary; therefore, the last 200 years is designated as a highstand systems tract. The stippled band indicates the tidal range, based on data from Chignecto Bay (Amos & Zaitlin, 1985; Amos *et al.*, 1991) and the Minas Basin (Dalrymple *et al.*, 1992a). Glacial ice occupied the area prior to 14 000 yr BP.

1980). Laminated marine silts accumulated in the Minas Basin at the same time (Amos, 1978). Based on the nature of the deposits, the tidal range is believed to have been microtidal (<2 m; Amos 1978; Wightman, 1980).

Following the highstand, relative sea level fell as isostatic uplift outpaced eustatic rise. A period of macrotidal conditions may have occurred during the fall (Fig. 3; Amos & Zaitlin, 1985; Amos *et al.*, 1991), based on the presence of coarse grained sediments and an erosion surface of this age in Chignecto Bay, and the similarity of basin geometry with the modern resonant configuration.

As the crest of the glacio-isostatic forebulge migrated NW through the area, a lowstand occurred in the inner Bay of Fundy from 5000–9000 yr BP (Fig. 3; Amos, 1978; Scott & Greenberg, 1983; Amos & Zaitlin, 1985). The minimum water level (at 7000–8000 yr BP) is not tightly constrained. In Chignecto Bay, submerged salt marsh peats have been recovered from depths as great as 25 m below present day, mean sea level (Scott & Greenberg, 1983; Amos & Zaitlin, 1985; Amos *et al.*, 1991), whereas salt marsh peats of this age lie only 10–15 m below present sea level at the west end of the Minas Basin (Fig. 4; Amos, 1978; Scott and Greenberg, 1983). The cause of the uneven elevation of the lowest shoreline is unknown, but possibilities include local differences in the height of the forebulge, recent differential subsidence or uplift, and later passage of the forebulge through the Minas Basin – Cobequid Bay area. The position of the lowstand shoreline in Cobequid Bay is unknown. It cannot be deeper than about −15 m (Fig. 4, maximum fall), but could be considerably shallower if the lowstand surface continues to rise to the east (Fig. 4, minimum fall).

During the late lowstand (Fig. 3), up to 8 m of silt and clay accumulated in the subtidal part of the Minas Basin (Amos, 1978). The tidal range is inferred to have been micro-mesotidal (2·0 ± 0·5 m; Amos,

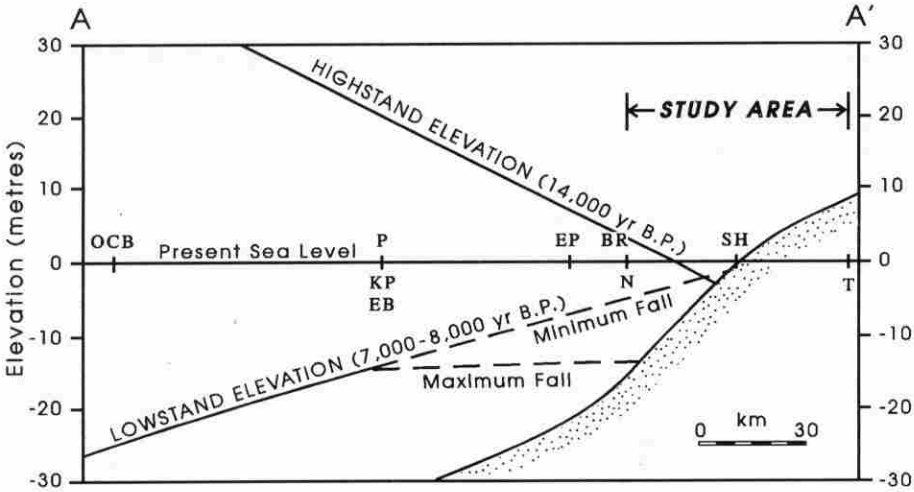


**Fig. 4.** East–west section along the length of Minas Basin and the CB-SR estuary (Fig. 1), showing the present day elevation of the shoreline at 14 000 yr BP (during the initial highstand; Fig. 3), and at 7000–8000 yr BP (during the late lowstand). The two dashed lines bracket the range of likely lowstand positions in Cobequid Bay. T, Truro; SH, Salter Head; N, Noel; BR, Bass River; EP, Economy Point; KP, Kingsport; EB, Evangeline Beach; P, Parrsboro; OCB, outer Chignecto Bay.

1978; Scott & Greenberg, 1983; Dalrymple *et al.*, 1992a). After 5000 yr BP, mean sea level rose steadily (Fig. 3; Scott & Greenberg, 1983; Amos & Zaitlin, 1985), as the glacial forebulge moved off to the NW and the area subsided. Tidal range increased simultaneously as the basin approached resonance with the lunar semi-diurnal ($M_2$) tide (Scott & Greenberg, 1983). Owing to the accompanying increase in tidal current speeds, the subtidal deposits in the Minas Basin for this period consist of a coarsening upward succession that passes from muds, through well sorted sand, into a surficial lag of shells, pebbles and cobbles. The modern sand bar – marsh complex in the CB-SR estuary (Fig. 2; Dalrymple *et al.*, 1990) accumulated during this time.

A recent compilation of data indicates that the tidal range increased rapidly from about 2 m at 4000 yr BP, to 9 m by 2800 yr BP, and thereafter more slowly to the present mean value of 12 m (Dalrymple *et al.*, 1992a). The $^{14}C$ dates obtained from basal peat deposits in Cobequid Bay salt marshes as part of this study indicate that the rate of rise of the high tide level has decreased from 25 cm/100 years at 2000 yr BP to approximately 10 cm/100 years at present. The history of the high tide elevation is significant because this represents base level in a tidal setting.

## STRATIGRAPHICAL OBSERVATIONS

In an earlier seismic survey of Cobequid Bay, Knight (1977, 1980) recognized two units overlying Triassic bedrock: Sediment 1, which consists of the modern tidal sediments; and Sediment 2, which was inferred to consist of Pleistocene till and/or glacial outwash. Examination of the seismic data collected in this study permits the distinction of three, informal units overlying Triassic bedrock (Figs 5 & 6). From the bottom up, these are referred to as Pleistocene, Lower Modern and Upper Modern. The lower two units represent a subdivision of Knight's (1977, 1980) Sediment 2, while the upper one is equivalent to his Sediment 1. The nature of the units and their bounding disconformities is described below; their sequence stratigraphical interpretation is discussed in the following section.

### Pleistocene unit

#### *Observations*

Throughout most of Cobequid Bay this unit has a sheet-like geometry, with a thickness typically <2 m (Figs 5 & 7). Its upper surface is marked by a high amplitude reflector, and no internal structure is visible. Near the west end of Cobequid Bay, the

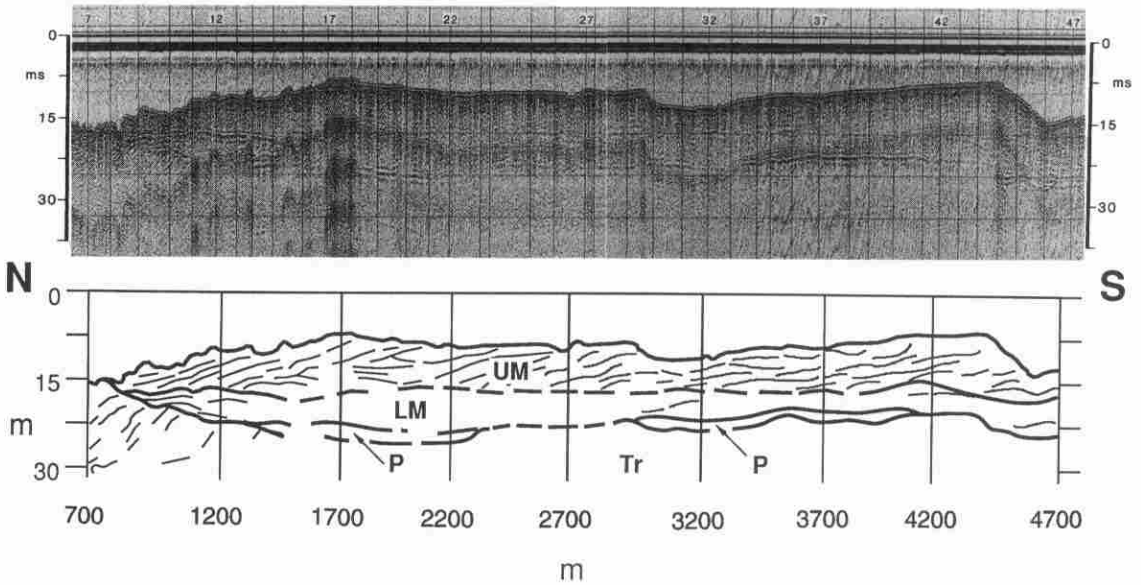


**Fig. 5.** Original record (upper) and interpretation (lower) of seismic section through the central portion of Cobequid Bay (see Fig. 2 for location) showing Triassic bedrock (Tr), a thin unit of Pleistocene (P) till; generally flat bedded, lagoonal deposits (central valley facies) of the Lower Modern (LM) unit (note onlap against northern valley wall); and tidal sand bars of the Upper Modern (UM) unit with prominent inclined stratification.

Pleistocene unit fills a 20 m deep depression cut into bedrock and older Pleistocene (?) sediments of unknown affinity (Fig. 6). Within the depression, the Pleistocene sediments consist of an acoustically massive lower portion that passes gradationally upward into sediments that contain moderately high amplitude, parallel, undulating reflectors. This bedded facies extends outside the depression, and comprises most of the Pleistocene unit in the western portion of Cobequid Bay. It is erosionally truncated and does not extend eastward into Cobequid Bay (Fig. 6). In the north-east, the Pleistocene unit forms a topographical high that rises 8 m above the valley floor (Fig. 7). Where reflectors are visible, stratification is either horizontal or dips gently to the south and west.

## Interpretation

Based on its stratigraphical position, this unit is inferred to be Pleistocene in age and to contain a variety of glacial and post-glacial sediments. The thin, sheet-like facies has been penetrated by cores near the margin of the bay, and consists of red sandy till that correlates with the Hants Till of Stea (1984). The depression at the western end of Cobequid Bay (Fig. 6; appears to be an eastward extension of a palaeovalley system that underlies the Minas Basin (Knight, 1977; Amos, 1978; Dalrymple *et al.*, 1992a). These valleys connect with the modern river valleys and were presumably cut by fluvial action prior to the latest glaciation. However, the apparent absence of pre-existing fluvial deposits within the valley, and the presence of localized, over-deepened depressions (Fig. 6; Hennigar, 1972) indicate that erosion by glacial ice and/or subglacial melt water has modified the original fluvial morphology. The massive sediments that fill the lower portion of the depression are interpreted to be outwash sands and gravels of subglacial or subaqueous origin, because relative sea level was too high at the time of deglaciation (Figs 3 & 4) to permit subaerial deposition.

The stratified material extending westward beyond the depression (Fig. 6), and the thick body of sediment in the north-east part of Cobequid Bay (Fig. 7), are believed to be deposits of immediate post-glacial age. The latter body is interpreted as an extension of the subaerial outwash plain that occupies the area near Glenholme (Fig. 2; Wightman, 1980) and the Salmon River valley at Truro (Hennigar, 1972). Sand and gravel pits contain deltaic forests and horizontal topset beds that presumably correspond with the

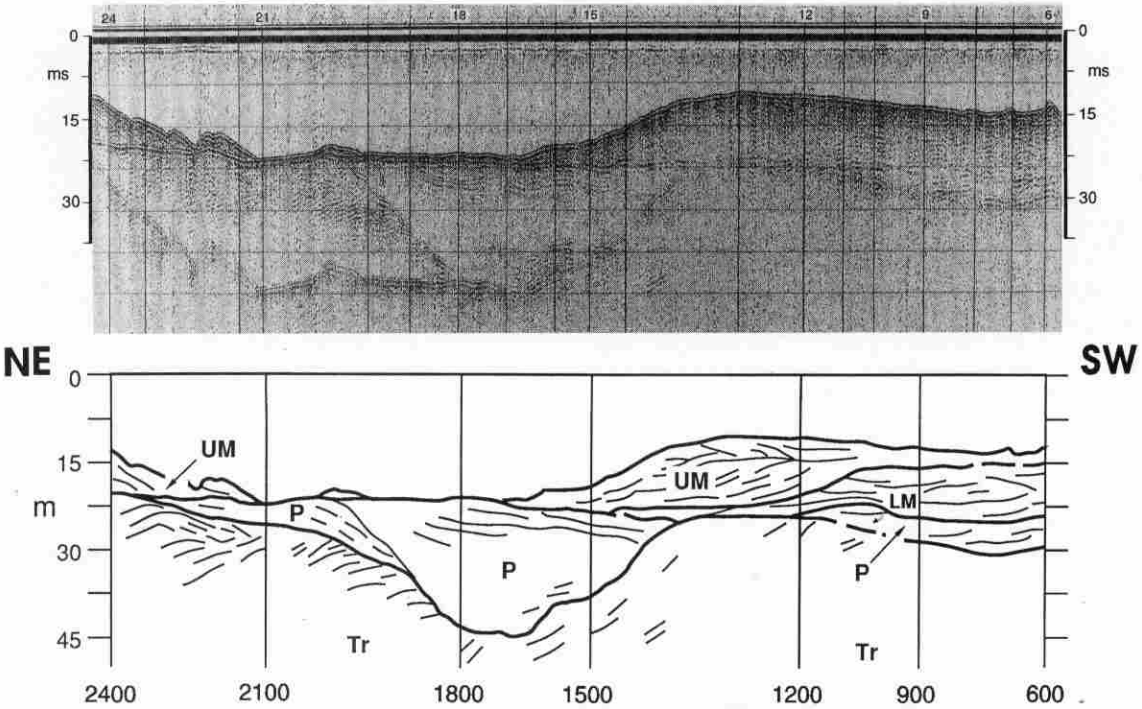


**Fig. 6.** Original record (upper) and interpretation (lower) of seismic section through the western portion of Cobequid Bay (see Fig. 2 for location) showing deep channel filled with Pleistocene (P) glacio-fluvial sediments. Stratified Pleistocene sediments at 600–1200 m are interpreted as glacio-marine outwash. Note the erosion of Pleistocene (P) and Lower Modern (LM) units at 1200 m.

structures seen in the seismic sections (Fig. 7). The upper unit of stratified sediments at the west end of Cobequid Bay (Fig. 6) is believed to consist of subtidal sands and/or muds which correlate with the glacio-marine muds that accumulated in the Minas Basin at this time (Amos, 1978). All of these stratified deposits formed when relative sea level was higher than at present (Figs 3 & 4).

## Lower Modern unit

### *Observations*

This unit overlies Triassic bedrock or Pleistocene sediment with a sharp contact. The topography of this surface (Fig. 8A) and the distribution of the Lower Modern unit (Fig. 8B) indicate that this unit occupies the lower portion of a broad valley sloping to the west. The thalweg is sinuous and bifurcates around a central high composed of Triassic bedrock. Significant erosion of underlying Pleistocene material is observed or inferred in only two areas: in the south-western corner of the survey area where up to 7–8 m of glacio-marine sediment has been removed (Fig. 6); and in the area just east of Salter Head where the Lower Modern sediments may occupy the seaward extension of the incised fluvial channel that was cut during the lowstand (see below). Lower Modern sediments locally reach 13 m in thickness, but are more typically about 5 m thick (Fig. 8B). Within this unit, four seismic facies have been identified (Fig. 8B).

*Western Ridge.* At the western end of Cobequid Bay, the Lower Modern unit occurs as a topographically positive ridge (Fig. 9) which rises up to 7·5 m above the top of the unit in adjacent areas. As seen in the isopach map (Fig. 8B), this ridge trends NW-SE across the valley. Its north-western end is not constrained by the seismic data, but the data show that it terminates before reaching the southern shore of the Bay. Thus, it is at least 3 km in length and 500–600 m wide.

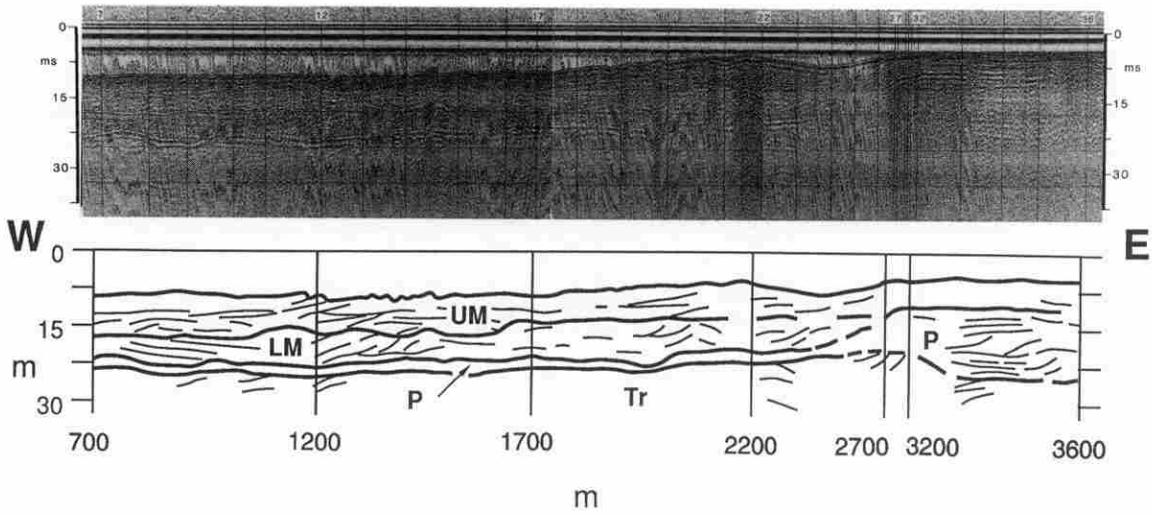


**Fig. 7.** Original record (upper) and interpretation (lower) of east–west seismic section showing Pleistocene (P) fluvio-deltaic deposits (2700–3600 m) in the north-central portion of Cobequid Bay (see Fig. 2 for location). The remainder of the Pleistocene unit is interpreted as a thin till sheet. In the Lower Modern (LM) unit, note the horizontal and westerly dipping stratification. They are interpreted as lagoonal sediments (central valley facies) and deltaic deposits (eastern clinoform facies), respectively.

Seismically, this ridge ranges from massive to faintly stratified. The visible reflectors are of low to modern amplitude and discontinuous. The majority dip gently (<1°) to the west, except along the eastern flank of the ridge where they dip eastward (Fig. 9). The sediments to the south and west of the ridge are horizontally stratified, with reflectors that are of slightly higher amplitude.

*Lobate Sand Body.* This facies lies to the north-east of the gap at the southern end of the Western Ridge. It is lobate in plan view, with dimensions of approximately 1 × 1 km. It lies within the thalweg of the pre-existing valley, and its top is elevated relative to its surroundings (Fig. 10); thus, it forms the thickest portion of the Lower Modern unit (Fig. 8B). Internally, it contains inclined, low amplitude reflectors that dip gently (0·5–1°) to the east, north and locally to the west. Because the seismic character of this facies is similar to that of the overlying macrotidal sand bars, it is inferred to consist of sand.

*Central Valley Facies.* To the north and east of the Lobate Sand Body, the Lower Modern unit is approximately 5 m thick (Fig. 8B) and has a relatively flat upper surface (Figs 5 & 7). Internal reflectors are not extensively developed but, where present, they are generally flat lying and onlap the valley walls. The amplitude of these reflectors is higher than in the modern tidal sand body. The implication is that this part of the Lower Modern unit contains greater acoustic contrast and, thus, greater lithological heterogeneity.

*Eastern Clinoforms.* Still further to the east, in the area north and west of Salter Head (Fig. 2), the Lower Modern unit comprises a number of sediment bodies which display westerly dipping internal reflectors (1–2°; Figs 7 & 11) of moderate amplitude. Each of these bodies is 3–5 m thick, and 0·3–1·5 km long in an east-west direction. In north-south sections, the reflectors are either flat lying or domed, suggesting that individual bodies have either a straight or lobate front. These clinoforms pass eastward into horizontal strata which are not well imaged through the overlying tidal sands.

### *Interpretation*

The following interpretations are based entirely on the isopach pattern, seismic character and stratal geometry of the facies because no cores have penetrated the Lower Modern unit.

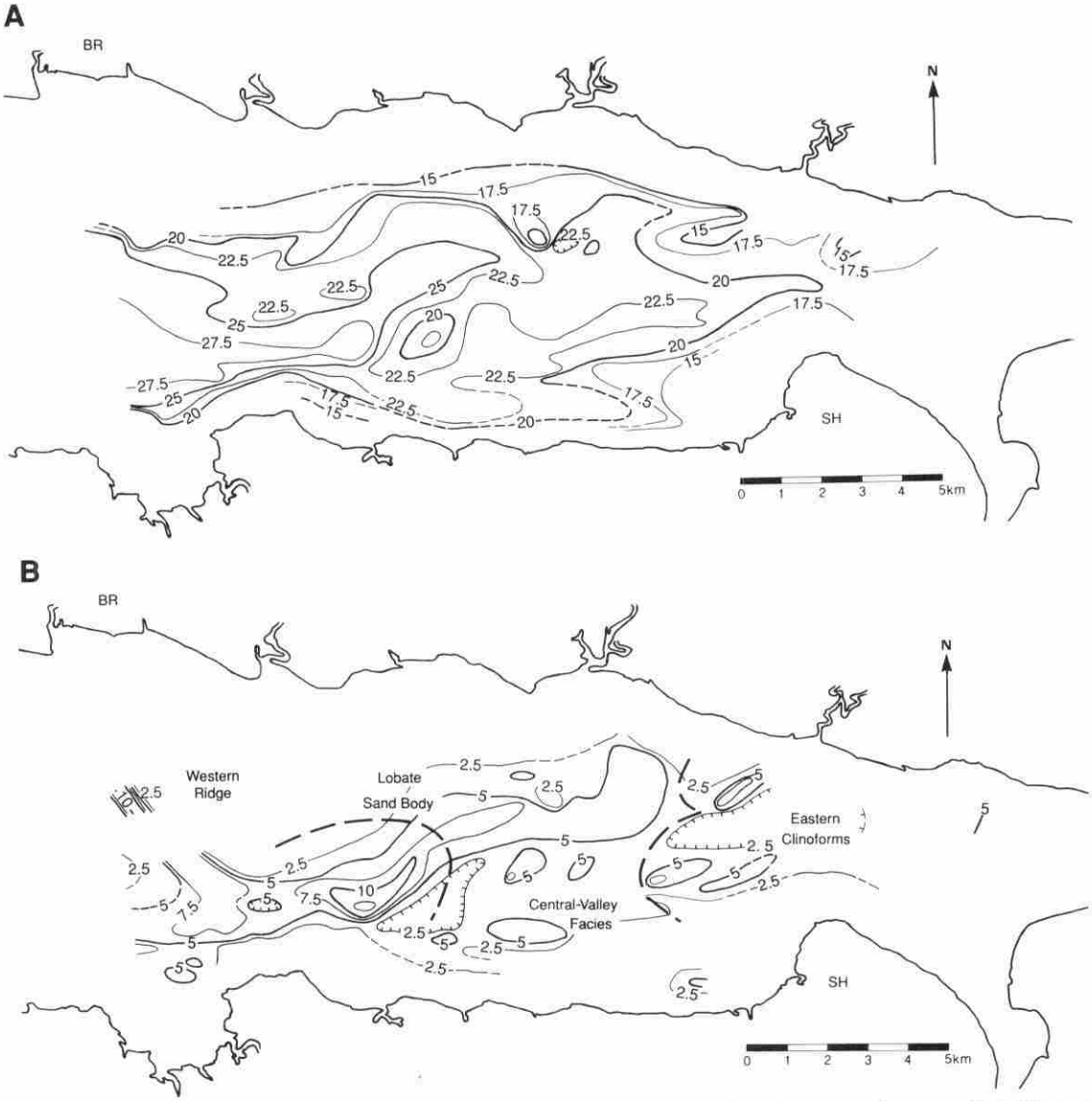


**Fig. 8.** (A) Topography of base of the Lower Modern unit, measured in metres below the present day, mean high tide level. (B) Isopach map of the Lower Modern unit (in metres) showing the distribution of the seismic facies described in the text. See Fig. 12 for the palaeogeographical interpretation. BR, Bass River; SH, Salter Head.

The Western Ridge is interpreted as a gravelly barrier spit (Fig. 12) which grew to the south-east from a bedrock headland at Bass River. The presence of gravel is inferred from the preserved positive relief of the ridge, despite the present day tidal currents, and its high acoustic reflectivity. The modern high tide beaches are also composed largely of gravel derived from till and outwash deposits (Dalrymple *et al.*, 1990). The westerly dipping reflectors presumably represent beach and/or shoreface deposits, while the easterly dipping reflectors on the east side of the ridge are washover deposits. The orientation of the ridge is consistent with the direction of longest fetch (Fig. 1), and of prevailing storm winds (Amos & Long, 1980).

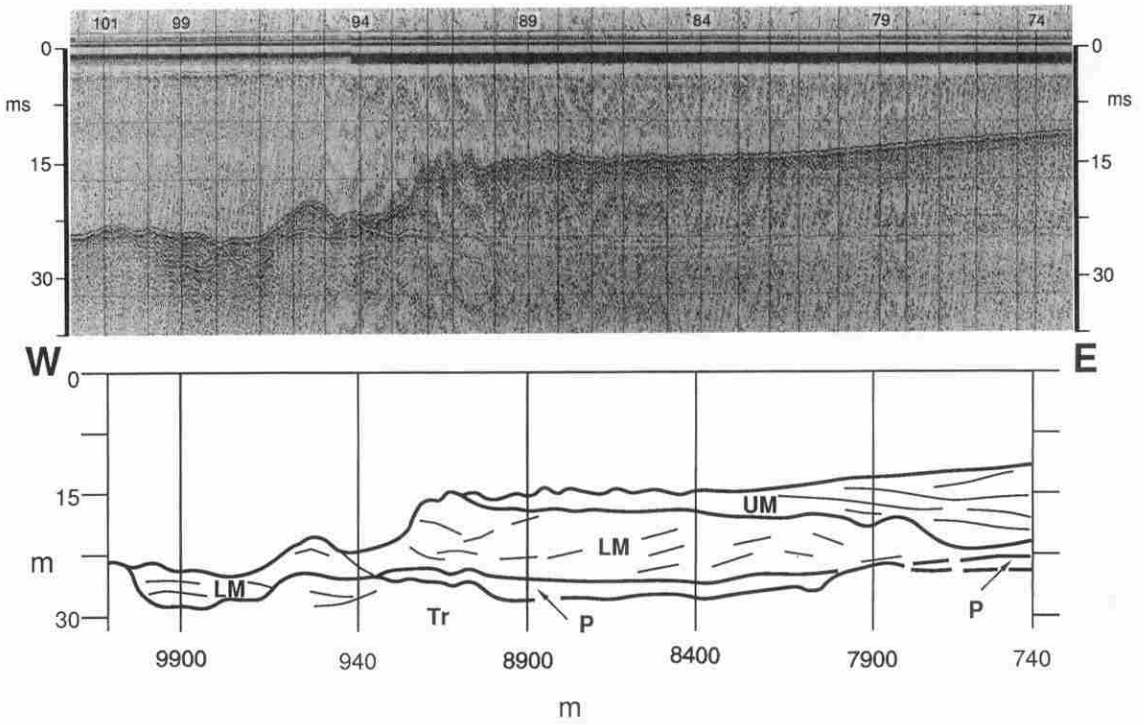


**Fig. 9.** Original record (upper) and interpretation (lower) of east–west seismic section through the Western Ridge, Lower Modern unit (LM), at the west end of Cobequid Bay (see Fig. 2 for location).

The gap in the ridge at its southern end is believed to a tidal inlet (Fig. 12). The flat bedded sediments in this area have reflectors which suggest that these inlet-associated sediments are sandy. Ebb tidal delta and inlet throat deposits may be present, but cannot be differentiated. Erosion of the underlying Pleistocene sediments in this area (Fig 6) is presumably due to tidal scour in the inlet.

The Lobate Sand Body is interpreted as a flood tidal delta which prograded into a back barrier lagoon, the finer grained deposits of which are represented by the Central Valley Facies (Fig 12). The Eastern Clinoforms are interpreted as bay head delta deposits, whereas the flat bedded sediments further to the east are inferred to be of fluvial origin. These latter deposits are likely to be coarse grained (sand and/or gravel) as the Salmon River is currently transporting gravel to the head of the estuary (Zaitlin, 1987; Dalrymple & Zaitlin, 1989).

The facies within the Lower Modern unit clearly indicate that deposition occurred within a wave-dominated estuary (Dalrymple *et al.*, 1992b); thus, conditions were considerably different from those in the present day estuary. The elevation of the barrier ridge and bay head delta relative to the sea level curve (Figs 3 & 4) suggests that deposition occurred 5000–9000 yr BP, during the lowstand or early part of the subsequent transgression. Independent evidence indicates that micro-mesotidal conditions prevailed throughout this time (Scott & Greenberg, 1983; Amos *et al.*, 1991; Dalrymple *et al.*, 1992a). Therefore, the tidal currents were apparently reduced to the point where wave action predominated (Davis & Hayes, 1984).

### Upper Modern unit

In the seismic sections from Cobequid Bay, the base of the modern, macrotidal sand body is generally marked by a strong, flat to gently undulating reflector with varying amounts of erosion of the underlying Lower Modern unit (Figs 5–7, 10 and 11). The topography of this surface (Fig. 13A) indicates that erosion is most pronounced along two headward terminating channels which occupy nearly the same locations as the main ebb- and flood-dominant

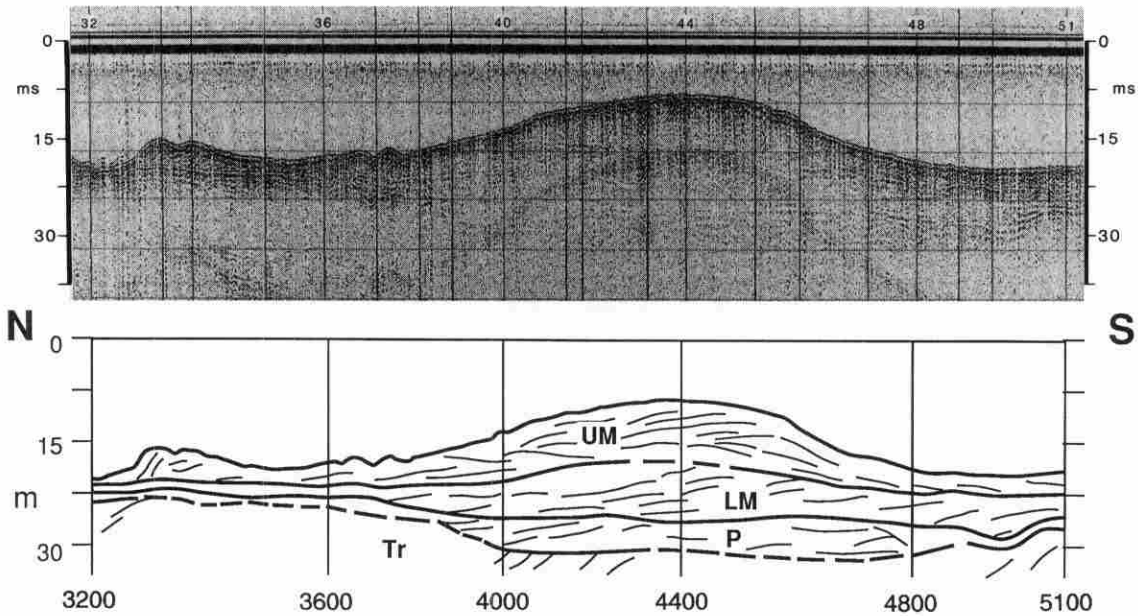


**Fig. 10.** Original record (upper) and interpretation (lower) of seismic section through the Lobate Sand Body (flood tidal delta; at 4000–4600 m) in the Lower Modern (LM) unit (see Fig. 2 for location).

channels in the modern estuary (Fig. 2). The greatest erosion occurs in the western part of Cobequid Bay, where up to 7 m of the Lower Modern unit has been removed (Fig. 6).

East of Salter Head, the modern tidal sediments overlie the westward dipping surface of the Pleistocene outwash, as well as the undulating till surface along the valley margins. A channel which was incised into the Pleistocene outwash before and/or during deposition of the lower Modern unit has been completely buried, except at the head of the estuary where infilling continues. Cores from the area east of Lyons Head (Fig. 2) show that this channel contains an unknown thickness of late Holocene (<2000 yr BP), fluvial sands and gravels which are distinguished from the overlying tidal sands by their coarser grain size, the absence of mud drapes and shell material, and the local presence of iron oxide staining. The top of these fluvial sediments lies as much as 4 m below the top of the Pleistocene outwash, and shows a rapid transition (<20 cm thick) into the tidal sands. The fluvial sediments become progressively older to the west, from <200 yr at the present location of the bayline (Fig. 2) to *c.* 2000 yr BP 5 km further seaward (Fig. 14). They are interpreted to be continuous with Lower Modern fluvial deposits (5000–9000 yr BP) which lie just east of Salter Head (Figs 12 and 14).

The modern tidal facies consists of an axial belt of sands which are flanked by mud flats and salt marshes in the area east of Salter Head (Figs 2 and 15; Dalrymple *et al.*, 1990, 1992b). The axial sands attain their greatest thickness (12 m) beneath the crests of the elongate bars at the western end of Cobequid Bay (Fig. 13B). From there, the sands thin to the east, and pinch out a short distance landward of the bayline (Fig. 2) where they interfinger with fluvial sand and gravel. None of the modern channels is floored by older material, except along the margins of facies zone 1 (Figs 5, 10 and 13B). In cores, the sand beneath the channels has the same characteristics throughout its thickness, suggesting that the sandy facies have aggraded vertically along the entire length of the estuary.

The seismic data reveal that the tidal sands have a complex architecture (Figs 5–7, 9–11). The sands of facies zone 1 are characterized by one or more superimposed sets of gently inclined (0·1–2·5°) stratification that generally dips either to the north or to the south. The strata in the uppermost set are commonly conformable with the side of the present bar

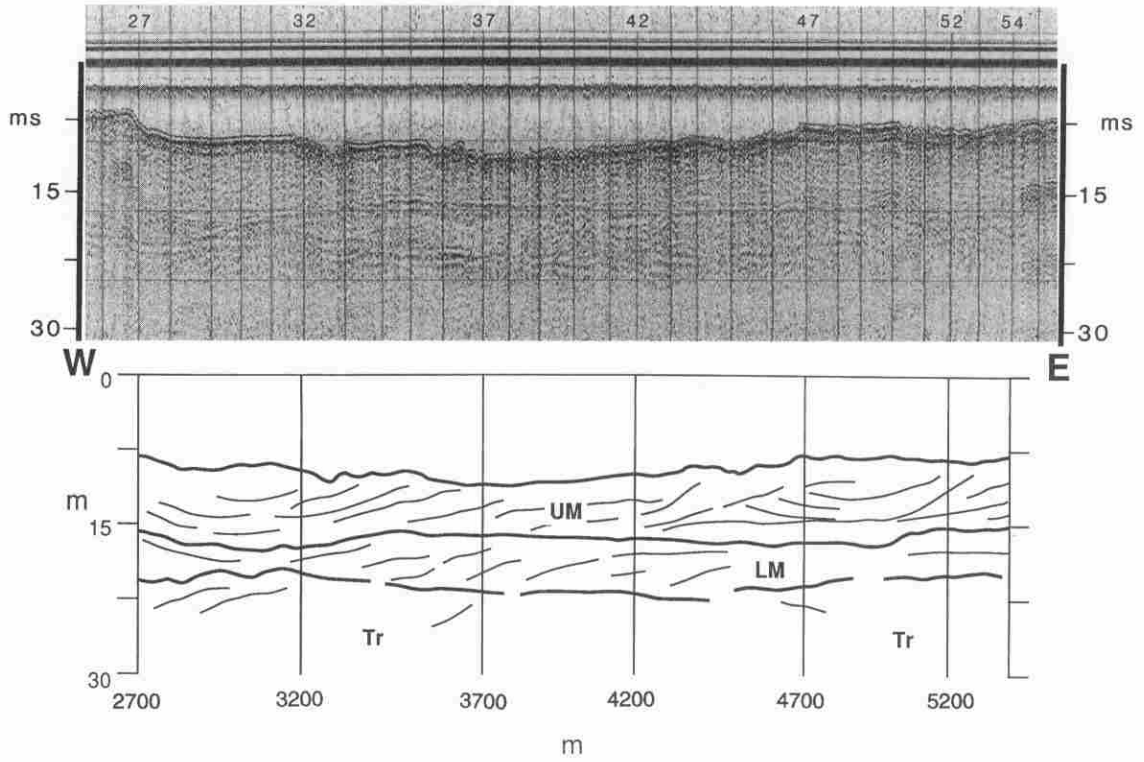


**Fig. 11.** Original record (upper) and interpretation (lower) of east–west seismic section through one of the Eastern Clinoform deltaic bodies in the Lower Modern (LM) unit (see Fig. 2 for location).

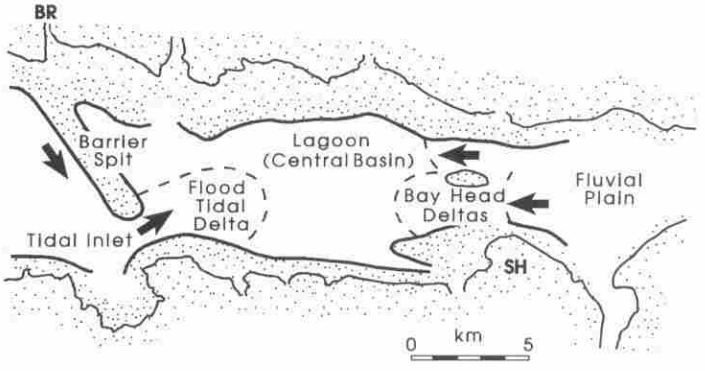


**Fig. 12.** Palaeogeography of the Lower Modern unit, based on the interpretation of the seismic facies: barrier spit = western ridge; flood tidal delta = lobate sand body; lagoon = central valley facies; and bay head deltas = eastern clinoforms. Arrows: the inferred direction of sediment movement. BR, Bass River; SH, Salter Head.

(Fig. 5), indicating that the bedding represents lateral accretion deposits. In facies zone 1, flat lying stratification is only present beneath the main ebb channel, but it occurs throughout the sand flats of facies zone 2 (Figs 2 and 14).

Cores collected in the tidal marshes (now mostly dyked) show that the flanking muddy deposits are locally more than 7 m thick, thinning outward toward the valley walls and eastward toward the tidal limit. In areas more than 100–500 m from the present

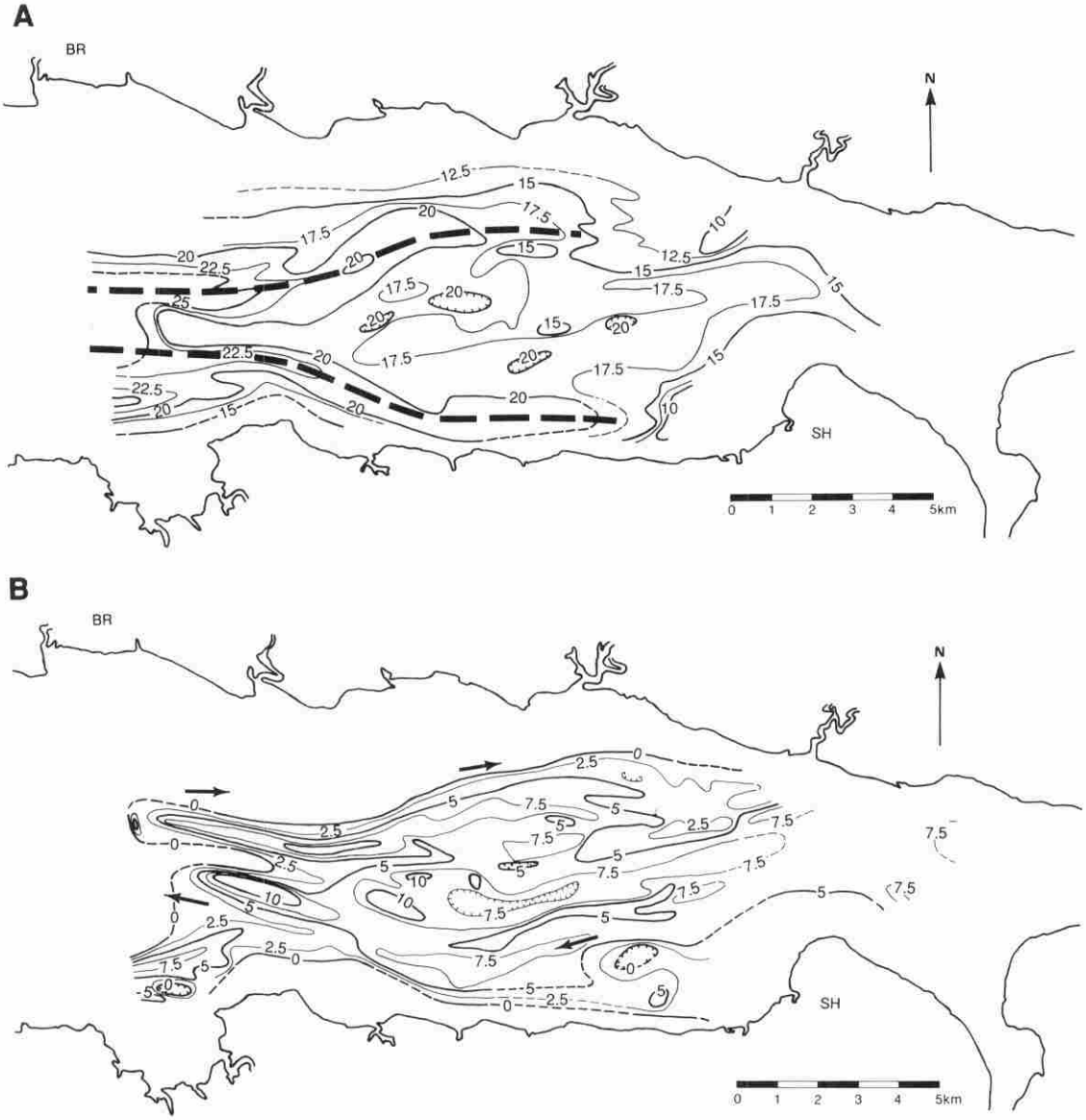


**Fig. 13.** (A) Topography of the base of the Upper Modern Sand Body, measured in metres below the present day, mean high tide level. The dashed lines indicate the thalwegs of the two erosional channels discussed in the text. This surface is a tidal ravinement surface. (B) Isopach map of the Upper Modern Sand Body (Fig. 2). The arrows indicate the direction of net sediment transport. BR, Bass River; SH, Salter Head.

channel margin, the marsh deposits consist of horizontal layers of organic-rich (10–50% organic material) and organic-poor silty mud which onlap the underlying till surface. Physical lamination is rare and roots are common to abundant throughout. Both sediment types are interpreted as gley (waterlogged) soils which accumulated in an anoxic marsh environment that aggraded slowly over the last 2200+ years (Fig. 15A), in response to rising high tide levels. The top of the till displays a bleached, podzolic soil profile that formed in a dry, upland setting, prior to passive burial by the marsh.

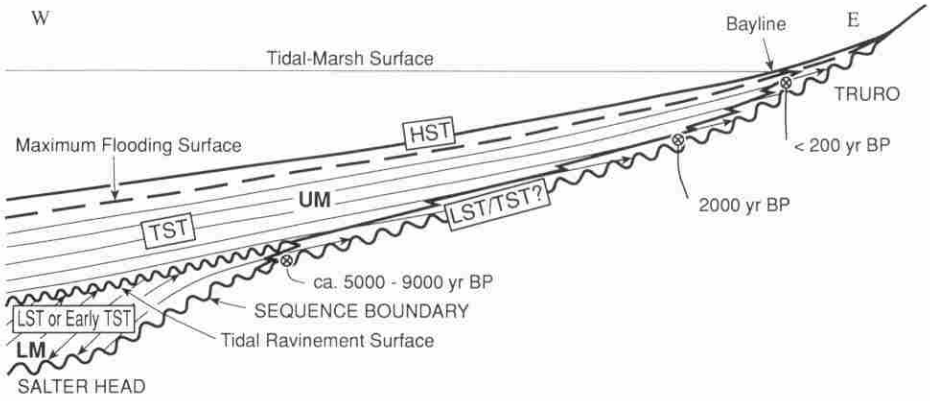


**Fig. 14.** Schematic section along the axis of the inner estuary (Salter Head to Truro), based on seismic data and cores. The key stratigraphical surfaces and interpreted systems tracts are shown; see Fig. 17 for a more complete section. Pleistocene outwash and till underlie the sequence boundary. The fluvial deposits (LST/TST?) are diachronous, extending continuously from the Lower Modern (LM) delta (5000–9000 yr BP) to the present day fluvial system landward of the bayline. See text for discussion of their systems tract assignment.

In contrast, the outermost parts of the marshes (Fig. 15A) along both sides of the estuary are underlain by a 6–7 m thick, fining upward succession that begins with parallel laminated, fine sand and passes upwards into thinly interbedded, silty muds and fine sands. These later sediments contain tidal rhythmites, neap-spring cycles and annual deposits similar to those described from the modern mudflats (Dalrymple *et al.*, 1991). Rooted marsh soils are present only at the surface, and are everywhere less than 100 years old (as determined by $^{14}C$ dating). Such successions are restricted to areas seaward of 250 year old, Acadian dykes (Fig. 15A), and represent a rapidly deposited, sand flat – mudflat – marsh succession which has accumulated recently, in front of an older, slowly aggrading marsh (Fig. 15A). Close core control indicates that the contact between these different successions is a steep, erosional scarp like those which occur at the outer edge of any present day marshes (Fig. 15A; Dalrymple *et al.*, 1991).

## DEPOSITIONAL HISTORY AND SEQUENCE STRATIGRAPHY

Depositional conditions within the Cobequid Bay – Salmon River valley have changed greatly throughout the post-glacial period. The significant reductions in sediment supply associated with removal of the Wisconsinan ice sheet, and the dramatic changes in tidal range (Fig. 3), have had a marked influence on the facies deposited within the valley. Simultaneous changes in relative sea level and the high tide elevation have had a significant influence on the stratal stacking patterns.

### Pleistocene highstand and sequence boundary formation

The sediments deposited during the 14 000 yr BP relative highstand (Fig. 3) represent the highstand systems tract (HST) of the first sequence preserved within the CB-SR valley. The HST consists of several prograding glacio-fluvial deltaic systems fed by outwash streams entering the head of Cobequid Bay at Truro and Glenholme (Figs 2, 16A and 17; Hennigar, 1972; Wightman, 1980). The contemporaneous subtidal sediments probably became finer westwards, from sands in Cobequid Bay, to muds in the Minas Basin (Amos, 1978). The lowstand (LST) and transgressive (TST) systems tracts of this sequence are not present in the study area because relative sea level was already high at the time of deglaciation.

The sequence boundary separating these sediments from the overlying Lower Modern unit (Fig. 17) varies in its expression along the length of the valley. In the area east of Lyons Head (Fig. 2), it coincides with the base of a narrow (50–250 m) incised channel which cuts the Pleistocene outwash plain (Fig. 16B). Within Cobequid Bay, the sequence boundary displays a broad, dendritic channel network (Fig. 8A). Erosion is inferred from the absence of glacio-marine sands from most of Cobequid Bay (the Lower Modern rests directly on till; Figs 9 and 16A,B).

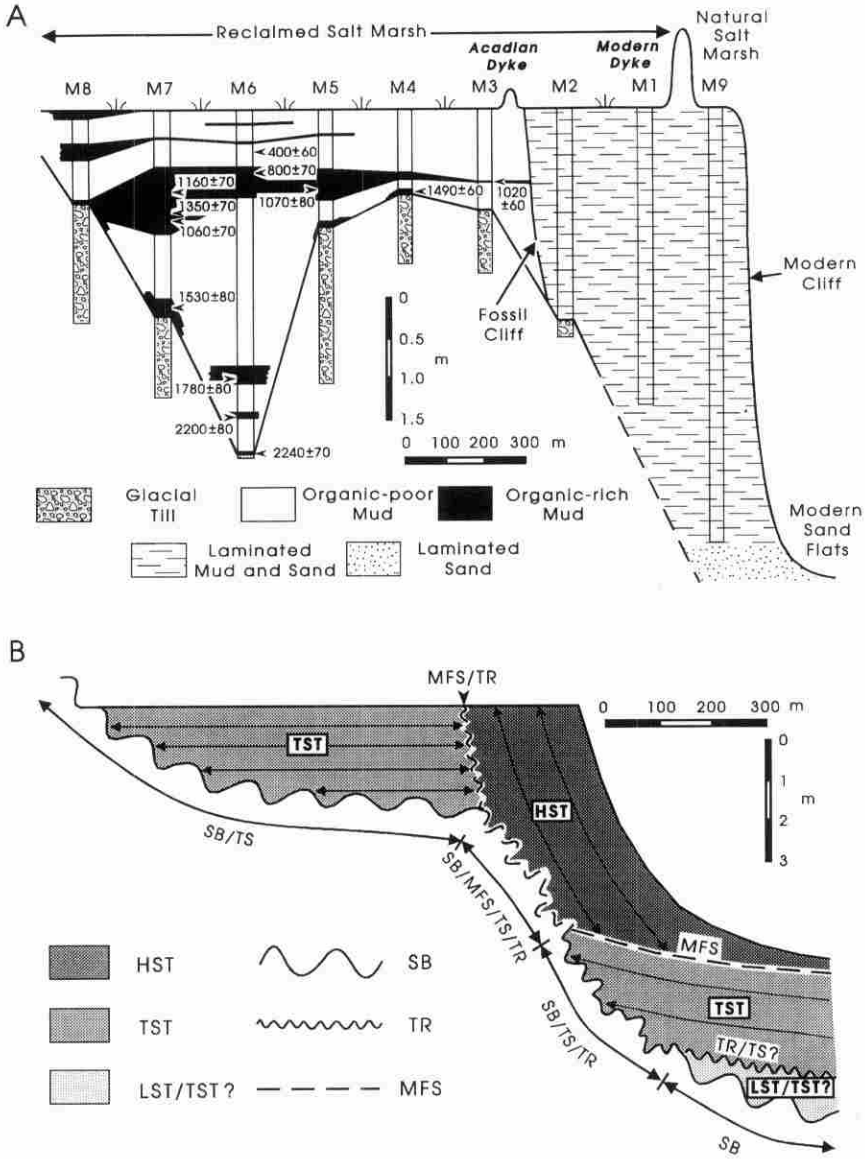


**Fig. 15.** (A) Stratigraphy of Masstown marsh, east of Glenholme (Fig. 2; M1–M9, core numbers). This section is at right angles to Fig. 14. The outer edge of the marsh is marked by a steep cliff that is formed by lateral migration of a tidal channel. The Acadian dyke was constructed prior to 1755, and borders a fossil cliff that separates slowly aggraded, organic-rich and organic-poor muds from laminated sands and muds (tidal rhythmites). (B) Sequence stratigraphical interpretation of (A), incorporating deeper sediments beneath the modern sand flats (note the difference in scales). Note the amalgamation and splitting of the surfaces. SB, sequence boundary; TS, transgressive surface; TR, tidal ravinement surface.

Further west in the Minas Basin, the surface corresponds with a pronounced hiatus, but no erosion has been noted (Amos, 1978). The narrow incised channel at the head of the estuary is clearly of fluvial origin. Incision may have commenced in response to the decrease in sediment discharge that accompanied the post-glacial reestablishment of vegetation, but probably continued throughout the sea level fall and

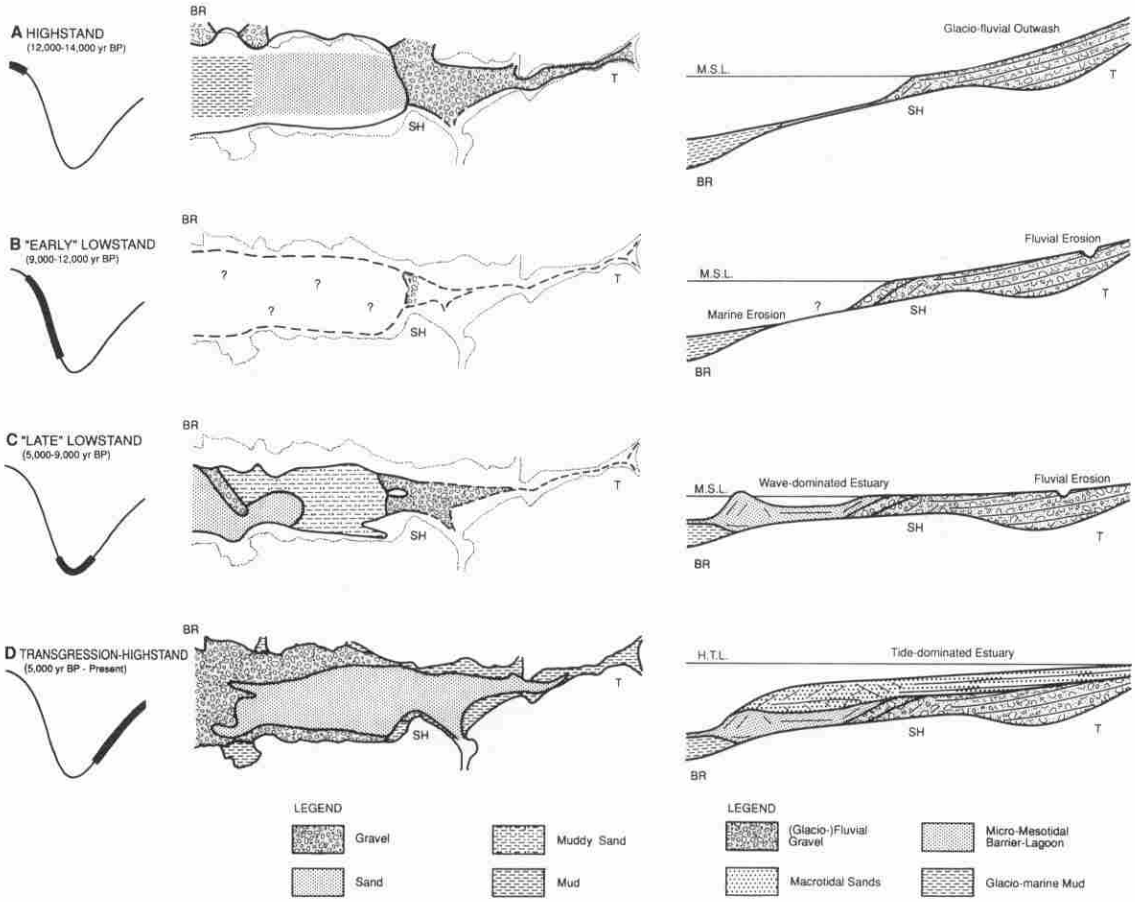


**Fig. 16.** Summary of the palaeogeographical and stratigraphical evolution of the CB-SR valley since deglaciation, assuming the minimum fall model (Figs 4 and 17B). (A) Highstand immediately following deglaciation. (B) Early lowstand. (C) Late lowstand (Fig. 12). (D) Transgression and present day highstand (Fig. 2). BR, Bass River; SH, Salter Head; T, Truro; M.S.L., mean sea level; H.T.L., high tide level.

into the subsequent transgression in headward areas (Fig. 16C). The westward extent of fluvial erosion is unknown, however, because of uncertainty regarding the lowstand shoreline position (Fig. 4). If the maximum fall occurred (Fig. 17A), fluvial erosion would have extended to the west of the study area. The dendritic channel which is present at the sequence boundary in Cobequid Bay is consistent with this possibility, but no Holocene fluvial deposits are recognized, and lowstand deltaic deposits have not been reported from the Minas Basin. If, instead, the minimum fall occurred, the lowstand shoreline would have remained within Cobequid Bay, perhaps located near Salter Head (Figs 4 and 17B), and some of the Eastern Clinoform deltas may have formed at this time (Fig. 16B). The erosional surface seaward of this would then be a 'regressive surface of erosion' (Posamentier *et al.*, 1992a) that was formed subtidally by wave and/or tidal current action during the sea level fall (Fig. 17B; Plint, 1988; Posamentier *et al.*, 1992b).

### Lowstand to early transgression (Lower Modern)

The assignment of the Lower Modern, wave-dominated estuarine complex to a systems tract depends on which of the lowstand positions is adopted. If the minimum fall is used (Fig. 17B), the Lower Modern unit represents the LST, and its upper surface is the transgressive surface (the flooding

surface separating the LST and TST; Posamentier & Vail, 1988; Van Wagoner *et al.*, 1988). On the other hand, if the maximum fall scenario is correct, the Lower Modern unit could be part of the TST, and the transgressive surface would be amalgamated with the underlying sequence boundary (Fig. 17A). Landward migration of the shoreline would have been accompanied by wave erosion and the formation of a wave ravinement surface seaward of the barrier. In either case, scour in the tidal inlet would have eroded older material (Fig. 12), producing a tidal ravinement surface (Fig. 17) that locally modified the sequence boundary.

**Late transgression to highstand (Upper Modern)**

Following deposition of the Lower Modern unit, the combination of rising mean sea level and simultaneous amplification of the tidal range (Fig. 3) produced the dramatic change from wave-dominated (Lower Modern) to tide-dominated (Upper Modern) deposition. At the same time, pronounced coastal onlap occurred; the bayline moved landward nearly 20 km, at a rate of up to $4{\cdot}5\ \mathrm{m\ yr^{-1}}$, while on the valley margins, the onlap of marsh deposits (Fig. 15) locally doubled the width of the valley floor.

Because tidal discharge increased with increasing tidal range, the macrotidal estuarine funnel (Dalrymple *et al.*, 1990) deepened, widened and shifted landward as the tidal limit transgressed. In the process, adjacent and underlying sediments were eroded by tidal currents (Fig. 18, stages 1–2); thus, the transgressing funnel was bounded by a tidal ravinement surface (Figs 15B and 17). On the floor of Cobequid Bay, this surface is marked by the incision of two tidal channels (Fig. 13A), and represents the flooding surface that separates the Lower and Upper Modern units. Along the margins of the outer bay (facies zone 1; Fig. 2), this surface coincides with the present day erosional foreshores (Dalrymple *et al.*, 1990). As a result of this ravinement, facies successions which record transgression of the macrotidal estuary are rare in the outer part of the area.

In the inner bay, the tidal ravinement surface is represented by the fossil cliff that separates laminated mudflat sediments from the older marsh deposits that had accumulated in the space between the estuarine funnel and the valley walls during the transgression (Figs 15A and 18; new and old marsh, respectively). In contrast, there appears to be little erosion in the axial portion of the inner estuary (facies zone 3; Fig. 2). Here, $^{14}C$ dates and observations in the modern valley show that fluvial sediments continued to accumulate at the head of the estuary throughout the transgression. As a result, the contact between Holocene fluvial gravels and the overlying estuarine sands is a diachronous facies boundary.

In recent work, it has become common to place the transgressive surface at the fluvial-estuarine contact (e.g. Posamentier *et al.*, 1992a; Allen & Posamentier, 1993). This may not be appropriate here, however, because this contact varies significantly in age along its length (Fig. 14), and is not a single physical surface. Its use as the transgressive surface would cause the LST and TST to overlap in age, and the TST would pinch out landward, causing the HST to lie directly on the LST in areas landward of the limit of estuarine sedimentation (Fig. 17A). This would violate the desired chronostratigraphical significance of systems tracts (Posamentier *et al.*, 1988; Van Wagoner *et al.*, 1988). Unfortunately, no other surface is presently recognized which would serve as an isochronous transgressive surface but, if one exists, it must lie within the Holocene fluvial deposits. The situation is less complicated beneath the fringing marshes. Here, the onlapping marsh deposits are clearly part of the TST, and the transgressive surface is coincident with the sequence boundary (Fig. 15B).

Throughout most of the estuary, *in situ* vertical aggradation characterizes the Upper Modern sediment body, both in the axial sands and the fringing marshes (Figs 14 and 15). This suggests that accumulation occurred at a time when sediment input approximately equalled the accommodation space created by the rise of the high tide elevation. Within the last 100–250 years, however, the estuary has entered a progradational phase (Dalrymple *et al.*, 1990), owing to a decrease in the rate of rise of the high tide elevation (Fig. 3), and/or to an increase in the rate of sediment supply (Amos & Long, 1980). This progradation is not yet evident in the axial sands, but is clearly shown by the presence of the rapidly-deposited, sand flat – mudflat – salt marsh succession along the outer margin of the fringing marshes (Figs 15 and 18, stages 2–3). These progradational deposits may be assigned provisionally to the HST (cf. Allen & Posamentier, 1993), whereas the vertically aggrading salt marsh deposits and axial sands are part of the TST (Figs 15B and 17). Thus, in these marginal areas, the maximum flooding surface (MFS) is represented by the fossil cliff which, as discussed above, is also the tidal ravinement surface. The MFS will be difficult to recognize in the axial sands, because it will be under- and overlain by similar facies.

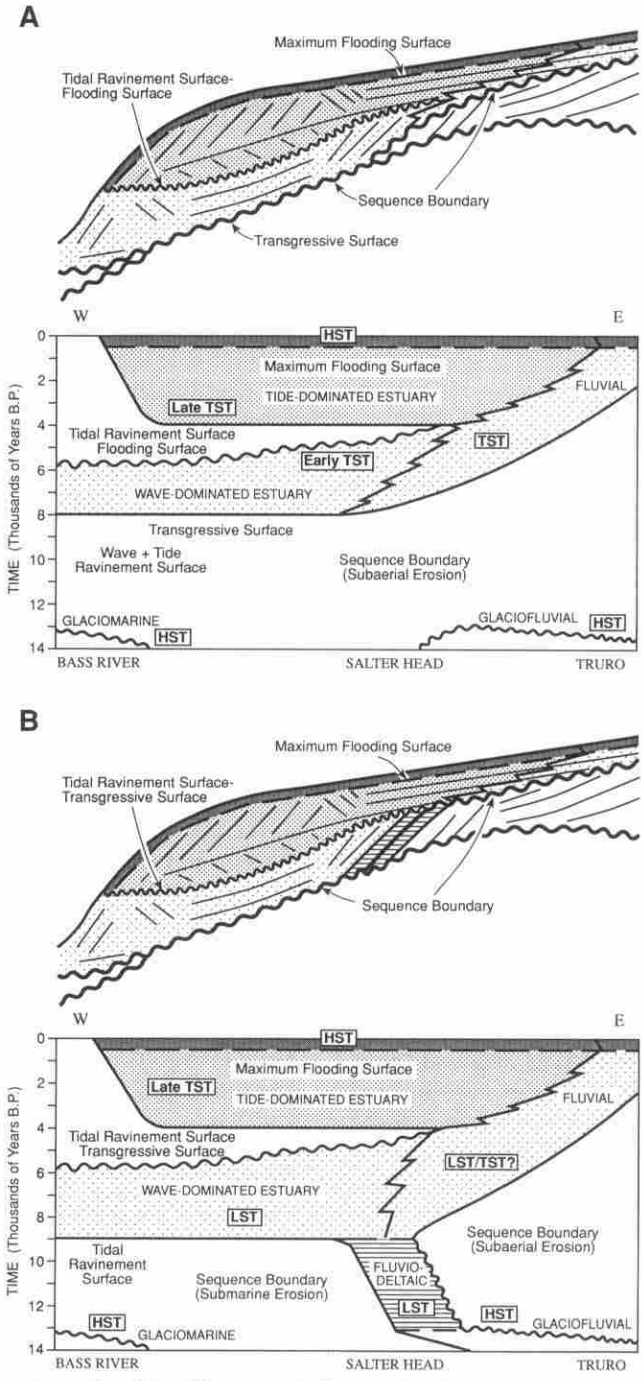


**Fig. 17.** Schematic section and stratigraphical (time–space) diagram showing systems tracts and key surfaces for (A) the maximum, and (B) the minimum sea level fall scenarios (Fig. 4). In (A), the lowstand shoreline lies to the west of the study area, whereas in (B), it lies near Salter Head (as shown in fig. 16B). In either interpretation, the positioning of the transgressive surface is uncertain in the area east of Salter Head (see text for discussion). Triassic bedrock underlies the lower sequence boundary in both sections.

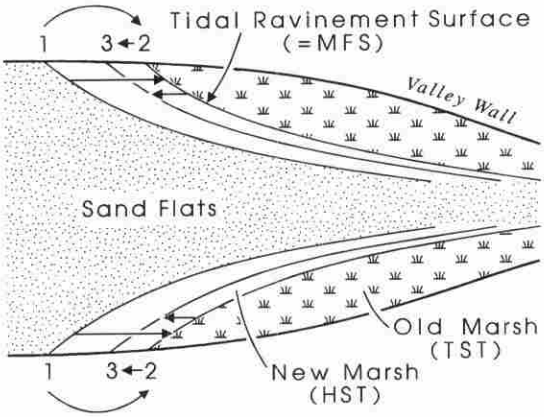


**Fig. 18.** Effects on the fringing marsh of transgression followed by regression. As the estuarine funnel (the area occupied by sand flats and mudflats) moves headward (1→2), old slowly aggrading salt marsh (part of the TST) is eroded by tidal channels or wave action (Allen, 1990; Dalrymple *et al.*, 1991), forming the tidal ravinement surface. As the estuarine funnel progrades seaward (2→3), new salt marsh is established on top of rapidly deposited mudflat sediments. This succession represents the beginning of the HST. The surface at 2 is the local expression of the MFS (Fig. 15).

## DISCUSSION AND COMPARISON WITH OTHER MACROTIDAL ESTUARIES

It is evident from the preceding that the fill of the Cobequid Bay – Salmon River valley contains two sequences: the lower one consists of Pleistocene outwash deposits, whereas the upper one contains two, retrogradationally stacked, estuarine parasequences. Accumulation of a third, progradational (highstand) unit began recently. Furthermore, because of the pronounced changes in environmental conditions over this period, the nature of the facies has changed markedly, from fluvially dominated deltaic, to wave-dominated estuarine and finally to tide-dominated estuarine. It is even more remarkable that all of this occurred in only 14 000 years. This illustrates how rapidly tidal range can change, due to changes in resonance, when the topography being transgressed is complex.

The only other macrotidal estuary for which a sequence stratigraphical analysis has been conducted is the Gironde (Allen & Truilhé, 1988; Allen, 1991; Allen & Posamentier, 1993). Because the Gironde estuary has experienced a monotonic sea level rise (as opposed to the fall and rise of the inner Bay of Fundy; Fig. 3), its fill contains only a single sequence; thus, the Gironde has a 'simple' fill, whereas the CB-ST valley has a 'compound' fill (Zaitlin *et al.*, 1994). In addition, parasequences comparable with those in the younger CB-SR sequence have not yet been recognized in the Gironde, perhaps because of differences in the rate of relative sea level rise. Despite these differences, a similar set of systems tracts can be recognized, because the two systems have evolved to approximately the same stage: both have an LST containing fluvial sediments, followed first by an onlapping TST consisting of estuarine deposits, and then by an HST composed of progradational estuarine and fluvial sediments. As a result, a similar set of surfaces can be identified in both systems. These similarities are consistent with the view that sequence stratigraphical principles are largely independent of scale and system-specific, environmental factors (Posamentier *et al.*, 1992a).

On the other hand, these environmental factors determine the nature of the facies and surfaces within the fill. Thus, the Gironde has been a mixed energy estuary (Dalrymple *et al.*, 1992b) throughout its history, with a well developed, tripartite facies distribution (wave built barrier/inlet complex; muddy central basin; and fluvially supplied but tide-dominated bay head delta; Allen & Posamentier, 1993). In comparison, conditions and the resulting facies organization have varied significantly in the CB-SR valley, ranging from wave-dominated during early Holocene time (Fig. 12), to strongly tide-dominated at present. The nature of the ravinement surface(s) is particularly influenced by the type of estuary. In mixed energy estuaries like the Gironde, two ravinement surfaces with distinctly different geometries are present (Allen & Posamentier, 1993; see also Demarest & Kraft, 1987): a lower, tidal ravinement surface formed by localized, deep scour in the tidal inlet; and an upper, regional, wave ravinement surface formed during landward passage of the shoreface. Tide-dominated estuaries like the modern CB-SR system differ in two significant ways. First, the wave ravinement surface is absent or poorly developed because a wave-dominated shoreface is not present. Second, the tidal ravinement surface is more widely developed, both laterally and longitudinally, because transgressive erosion extends throughout most of the estuary, rather than being localized near the inlet. Despite this, the tidal ravinement surface may not incise as deeply as in mixed energy estuaries (maximum incision 7 m in Cobequid Bay, versus 25 m in the Gironde), due to the lack of a wave built barrier and significant flow constriction.

In the Gironde estuary, the MFS is characterized by *seaward directed* downlap of bay head delta sand bars onto central basin muds (Allen & Posamentier, 1993). In a tide-dominated estuary such as the CB-SR system, there is no bayhead delta (Dalrymple *et al.*, 1992b). Instead, the MFS is associated with *inward directed* progradation of mudflats (Fig. 15B) from both sides. In both cases, the MFS will be difficult to recognize because a condensed section, such as marks the MFS in shelf sediments, is not present.

The timing of fluvial deposition is apparently different in the two estuaries. In the Gironde, most of the fluvial sediment is believed to have accumulated during the late lowstand (little or no fluvial deposition occurred during the transgression), whereas in the CB-SR estuary, fluvial deposition was continuous throughout the transgression, with the locus of sedimentation migrating landward as the bayline transgressed (Fig. 14). Such differences are to be expected, given that the amount of fluvial accumulation at any location depends on the relative rates of sediment supply and transgression. If there is no fluvial deposition during the transgression, the fluvial-estuarine facies boundary and transgressive surface are coincident, as interpreted by Allen & Posamentier (1993). Wright & Marriott (1993) argue, however, that fluvial aggradation is to be expected during the transgression. Therefore, if the transgressive surface is defined as a time horizon separating aggradational/progradational LST deposits from the retrogradational TST, it may more commonly lie below the fluvial-estuarine contact. Whether this change in stacking occurs at a discrete, easily recognized surface is uncertain, and further research is needed to define criteria for distinguishing the LST and TST in incised valley successions.

Although less is known about the sequence stratigraphy of other macrotidal estuaries, some additional comparative observations are possible. The Severn estuary appears not to have reached the point of maximum transgression, and the margins of the estuarine funnel continue to erode (Allen, 1990). The result will be a surface with a geometry similar to that of the tidal ravinement surface in Cobequid Bay. In contrast, the South Alligator River is now depositing a progradational HST that overlies transgressive deposits (Woodroffe *et al.*, 1989, 1993); the MFS corresponds with the big swamp period. LST deposits are not reported, and no information exists on the nature of any ravinement surfaces, but the Holocene valley fill is simple (only one sequence is present).

Finally, Chignecto Bay, Bay of Fundy (Fig. 1), has a similar sea level and tidal range history to Cobequid Bay (Amos & Zaitlin, 1985; Amos *et al.*, 1991), and contains a compound fill that is broadly comparable with that described above. Unlike Cobequid Bay, however, the upper sequence contains three, retrogradationally staked parasequences, as opposed to the two observed in the CB-SR study area. This difference may be due to the fact that the magnitude of the sea level rise and the amount of accommodation space created were greater in Chignecto Bay (Fig. 4). Alternatively, if the maximum fall scenario applies in Cobequid Bay (Figs 4 and 17A), then a third, older parasequence may exist further seaward, outside the study area.

## SUMMARY AND CONCLUSIONS

The post-glacial sediments within the Cobequid Bay – Salmon River incised valley contain two sequences, the upper one still in the process of formation. The older sequence formed during deglaciation, and consists of alluvial outwash, glaciomarine deltas and associated subtidal sediments which are all assigned to the HST (Fig. 17); the LST and TST are not present in the study area because relative sea level was at its highest point when ice left the area. The subsequent sea level fall led to the production of a sequence boundary, with fluvial erosion in subaerial parts of the valley, and perhaps also wave erosion in areas seaward of the lowstand shoreline. It was later modified by wave action (?) and local tidal scour during the early transgression.

The upper sequence contains two retrogradationally stacked parasequences which are separated by a tidal ravinement surface that extends throughout much of the modern estuary. The lower unit, which is part of the LST or early TST, accumulated in a wave-dominated estuarine setting under micromesotidal conditions. It possesses the tripartite facies subdivision which typifies such estuaries: a marine sediment body consisting of a barrier spit, tidal inlet and flood tidal delta; a finer grained central basin; and a fluvially dominated bay head delta. The upper parasequence formed under the modern macrotidal conditions, and displays *in situ* aggradation. It represents the late TST. In very recent time, progradation has begun, signalling the beginning of the HST. The MFS is easily identified only along the margins of the inner estuary, where it coincides with the tidal ravinement surface; in the axial sandy facies it is likely to be

a cryptic surface. Fluvial deposition continued throughout the transgression, causing the fluvial-estuarine facies contact to be diachronous.

The following general conclusions can be derived from these observations.

1. Incised valley fills can be very complex, and it is possible to identify parasequences, parasequence sets and sequences within an individual valley. This is particularly so if the valley is not filled in a single depositional interval and more than one sequence is present (i.e. the fill is compound as opposed to simple). Nevertheless, careful use of sequence stratigraphical concepts allows definition of a succession of surfaces and systems tracts within high frequency, incised valley sequences that mimic those in lower frequency, shelf sequences (e.g. Posamentier & Vail, 1988). Stratigraphical differences between modern macrotidal estuaries are consistent with differences in their relative sea level history, although changes in high tide elevation due to variations in tidal range have a complicating influence.
2. Local environmental factors have a significant influence on the characteristics of the various surfaces. The distinction between wave- and tide-dominated estuaries (Dalrymple *et al.*, 1992b) is particularly significant with respect to the nature of the ravinement surfaces. Differences may even exist between adjacent systems, because of differences in sediment supply relative to the accommodation space (as partially influenced by valley depth).
3. The nature of the stratigraphical surfaces (sequences boundary, ravinement surfaces and flooding surfaces) may differ significantly from place to place within the valley fill. Local amalgamation of surfaces is common.
4. As noted by Allen & Posamentier (1993), the tidal ravinement surface is a significant feature within incised valley fills. It is most extensively developed in tide-dominated estuaries, but incision may be greatest in mixed energy estuaries because they have both a constriction at the mouth and a large tidal prism.
5. Because fluvial deposition can continue throughout the transgression, the fluvial-estuarine contact may be diachronous and lie above transgressive deposits; thus, this contact may not be an appropriate choice for the transgressive surface. The transgressive surface is more appropriately placed where stacking patterns become retrogradational, which may occur within the basal fluvial portion of the valley fill.
6. The transgression of irregular, highly embayed topography can cause geologically instantaneous changes in tidal range. This, in turn will have a profound impact on the nature of the facies and surfaces, and may lead to the development of discrete parasequences.

## ACKNOWLEDGMENTS

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